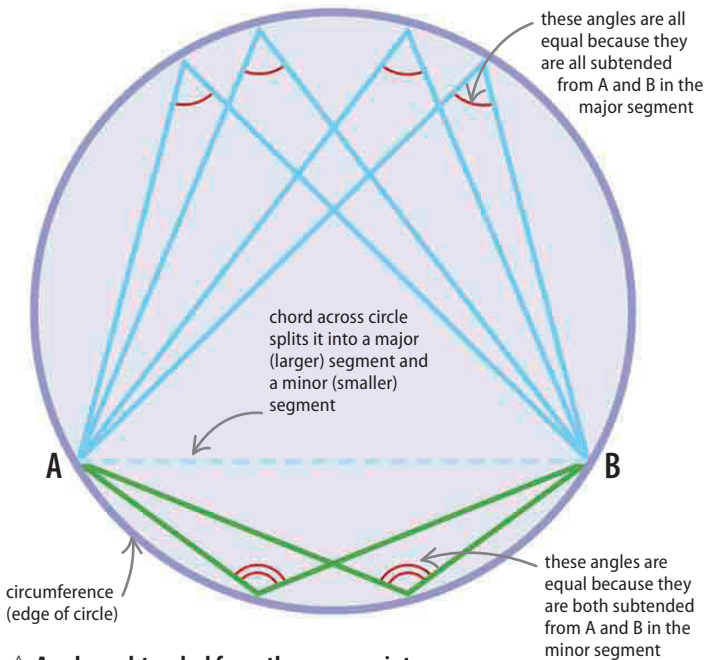
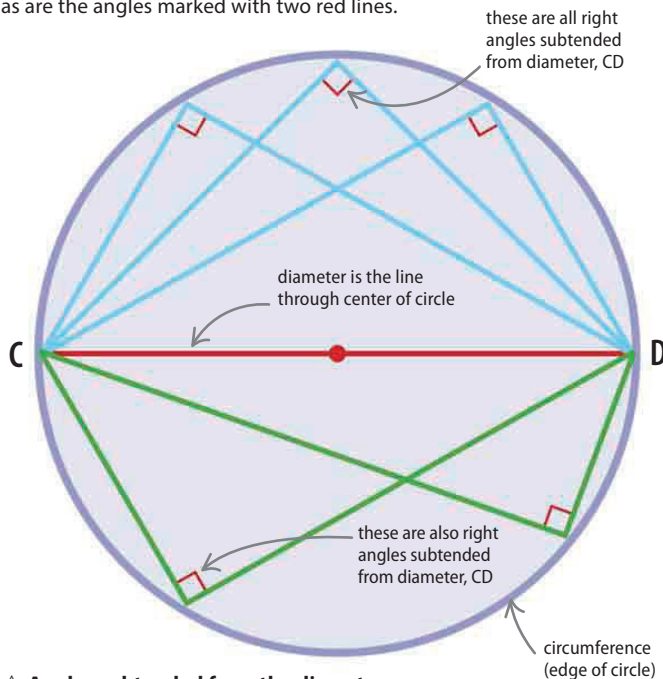




△ Angles subtended from the same points

Angles at the circumference subtended from the same two points in the same segment are equal. Here the angles marked with one red line are equal, as are the angles marked with two red lines.



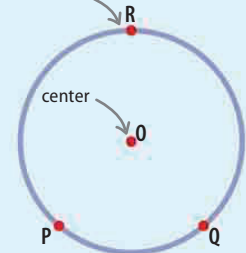
△ Angles subtended from the diameter

Any angle at the circumference that is subtended from two points either side of the diameter is equal to 90° , which is a right angle.

Proving angle rules in circles

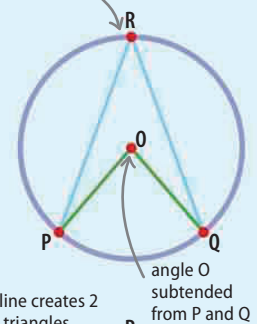
Mathematical rules can be used to prove that the angle at the center of a circle is twice the size of the angle at the circumference when both the angles are subtended from the same points.

R, P, and Q are 3 points on the circumference



Draw a circle and mark any 3 points on its circumference, for example, P, Q, and R. Mark the center of the circle, in this example it is O.

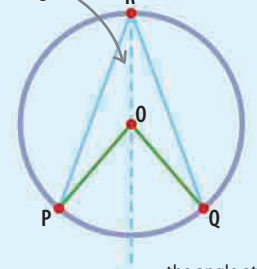
angle R subtended from P and Q



Draw straight lines

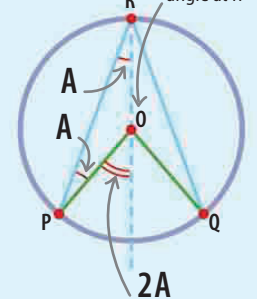
from R to P, R to Q, O to P, and O to Q. This creates two angles, one at R (the circumference of the circle) and one at O (the center of the circle). Both are subtended from points P and Q.

dividing line creates 2 isosceles triangles



Draw a line from R through O, to the other side of the circle. This dividing line creates two isosceles triangles. Isosceles triangles have 2 sides and 2 angles that are the same. In this case, two sides of triangles POR and QOR are formed from 2 radii of the circle.

the angle at O is twice the angle at R



For one triangle the two angles on its base are equal, and labeled A. The exterior angle of this triangle is the sum of the opposite interior angles (A and A), or $2A$. Looking at both triangles, it is clear that the angle at O (the center) is twice the angle at R (the circumference).